

Redes Neuronales

U2 Redes Simples

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Section 1

Perceptron Learning Rule Convergence Theorem

Perceptron learning rule

The **perceptron learning rule** is a follow:

- Given a finite set of P input training vectors

$$x(p), \quad p = 1, \dots, P$$

- Each with an associated target value

$$t(p), \quad p = 1, \dots, P$$

which is either $+1$ or -1

Perceptron learning rule

- An activation function $y = f(y_{in})$ where

$$y = \begin{cases} 1 & \text{if } y_{in} > \theta \\ 0 & \text{if } -\theta \leq y_{in} \leq \theta \\ -1 & \text{if } y_{in} < -\theta \end{cases}$$

- The weights are updated as follows:

- If $y \neq t$ then

$$w(new) = w(old) + tx$$

- Else

no change in the weights

Perceptron learning rule convergence theorem

The **perceptron learning rule convergence theorem** is:

If there is a weight vector w^* such that

$$f(x(p) \cdot w^*) = t(p), \quad \forall p$$

then for any starting vector w , the perceptron learning rule will converge to a weight vector (**not necessarily unique and not necessarily** w^*) that gives the correct response for all training patterns, and it will do so in a **finite number of step**.

Proof

The proof of the theorem is simplified by the observation that the training set can be considered to consist of two parts:

$$F^+ = \{x \text{ such that the target value is } +1\}$$

$$F^- = \{x \text{ such that the target value is } -1\}$$

A new training set is then defined as

$$F = F^+ \cup -F^-$$

where

$$-F^- = \{-x \text{ such that } x \text{ is in } F^-\}$$

In order to simplify the algebra slightly, we shall assume, without loss of generality, that $\theta = 0$ and $\alpha = 1$ in the proof.

The existence of a solution of the original problem, namely the existence of a weight vector w^* for which

$$x \cdot w^* > 0 \quad \text{if } x \text{ is in } F^+$$

$$x \cdot w^* < 0 \quad \text{if } x \text{ is in } F^-$$

is equivalent to the existence of a weight vector w^* for which

$$x \cdot w^* > 0 \quad \text{if } x \text{ is in } F$$

All the target values for the modified training set are +1.

If the response of the net is incorrect for a given training input, the weights are updated according to

$$w(new) = w(old) + x$$

Note that the input training vectors must each have an additional component (which is always 1) included to account for the signal to the **bias weight**.

We assume that the training set has been modified so that all targets are $+1$.

Note that this will involve reversing the sign of all components (including the input component corresponding to the bias) for any input vectors for which the target was originally -1 .

We now consider the sequence of input training vectors for which a weight change occurs. We must show that this sequence is **finite**.

Let the starting weights be denoted by $w(0)$, the first new weights by $w(1)$, etc.

If $x(0)$ is the first training vector for which an error has occurred, then

$$w(1) = w(0) + x(0), \quad \text{where by assumption } x(0) \cdot w(0) \leq 0$$

If another error occurs, we denote the vector $x(1)$; this vector may be the same as $x(0)$ if not errors have occurred for any other training vectors, or $x(1)$ may be different from $x(0)$. In either case,

$$w(2) = w(1) + x(1) \quad \text{where by assumption } x(1) \cdot w(1) \leq 0$$

At any stage, say, k , of the process, the weights are changed if and only if the current weights fail to produce the correct (positive) response for the current input vector, i.e., if $x(k-1) \cdot w(k-1) \leq 0$.

Combining the successive weight changes gives

$$w(k) = w(0) + x(0) + x(1) + \dots + x(k-1)$$

We now show that k cannot be arbitrarily large. Let w^* be a weight vector such that $x \cdot w^* > 0$ for all training vectors in F .

Let $m = \min_{x \in F} \{x \cdot w^*\}$, this minimum exists whenever there are only finitely many training vectors. now,

$$\begin{aligned} w(k) \cdot w^* &= [w(0) + x(0) + x(1) + x(2) + \dots + x(k-1)] \cdot w^* \\ &\geq w(0) \cdot w^* + km \end{aligned}$$

since $x(i) \cdot w^* \geq m$ for each i , $1 \leq i \leq P$.

The Cauchy-Schwartz inequality states that for any vectors a and b

$$\begin{aligned}(a \cdot b)^2 &\leq \|a\|^2 \|b\|^2 \\ \|a\|^2 &\geq \frac{(a \cdot b)^2}{\|b\|^2}, \quad \text{for } \|b\|^2 \neq 0\end{aligned}$$

Therefore,

$$\begin{aligned}\|w(k)\|^2 &\geq \frac{(w(k) \cdot w^*)^2}{\|w^*\|^2} \\ &\geq \frac{(w(0) \cdot w^* + km)^2}{\|w^*\|^2}\end{aligned}$$

This shows that the squared length of the weight vector grows faster than k^2 , where k is the number of times the weights have changed.

However, to show that the length cannot continue to grow indefinitely, consider

$$w(k) = w(k-1) + x(k-1)$$

together with the fact that

$$x(k-1) \cdot w(k-1) \leq 0$$

By simple algebra¹,

$$\begin{aligned} \|w(k)\|^2 &= \|w(k-1)\|^2 + 2x(k-1) \cdot w(k-1) + \|x(k-1)\|^2 \\ &\leq \|w(k-1)\|^2 + \|x(k-1)\|^2 \end{aligned}$$

¹Why? (tip: $\|z\|^2 = z \cdot z$)

Now let $M = \max_{x \in F} \{\|x\|^2\}$, then

$$\begin{aligned}
 \|w(k)\|^2 &\leq \|w(k-1)\|^2 + \|x(k-1)\|^2 \\
 &\leq \|w(k-2)\|^2 + \|x(k-2)\|^2 + \|x(k-1)\|^2 \\
 &\vdots \\
 &\leq \|w(0)\|^2 + \|x(0)\|^2 + \dots + \|x(k-1)\|^2 \\
 &\leq \|w(0)\|^2 + kM
 \end{aligned}$$

Thus, the squared length grows less rapidly than linearly in k .

Combining the inequalities

$$\begin{aligned} \|w(k)\|^2 &\geq \frac{(w(0) \cdot w^* + km)^2}{\|w^*\|^2} \\ \|w(0)\|^2 + kM &\geq \|w(k)\|^2 \end{aligned}$$

shows that the number of times that the weights may change is bounded. Specifically,

$$\frac{(w(0) \cdot w^* + km)^2}{\|w^*\|^2} \leq \|w(k)\|^2 \leq \|w(0)\|^2 + kM$$

Again, to simplify the algebra, assume (without loss of generality) that $w(0) = 0$.

Then the maximum possible number of times the weights may change is given by

$$\frac{(km)^2}{||w^*||^2} \leq kM$$
$$k \leq \frac{M||w^*||^2}{m^2}$$

Since the assumption that w^* exists can be restated, without loss of generality, as the assumption that there is a solution weight vector of unit length (and the definition of m is modified accordingly), the maximum number of weight update is

$$\frac{M}{m^2}$$

Note, however, that many more computations may be required, since very few input vectors may generate an error during any one epoch of training.

Also, since w^* is unknown (and therefore so is m), the number of weight updates cannot be predicted from the preceding inequality.

References

- Fausset, L. *Fundamentals of Neural Networks. Architectures, Algorithms and Applications*, Pearson Education, 1994.